

PROTECTION AND CO-ORDINATION STUDY

1. Nominal Equipment Data

Equipment Description	Nominal Values
Transformer Dielectric Medium	SECO
Nominal Power Rating	1000 kVA
Percentage Impedance (Z%)	5.0 %
Voltages (Primary / Secondary)	13.8 kV / 0.38 kV

2. Currents and Limiting Factors

Current Parameter	Result
Nominal Phase Current	50.26 A
Short-Circuit Current (Icc3)	837.73 A
Inrush Current (Peak / Time)	586.41 A / 0.10s
Calculated Accuracy Limit Factor (ALF)	13.96

3. CT Saturation Analysis

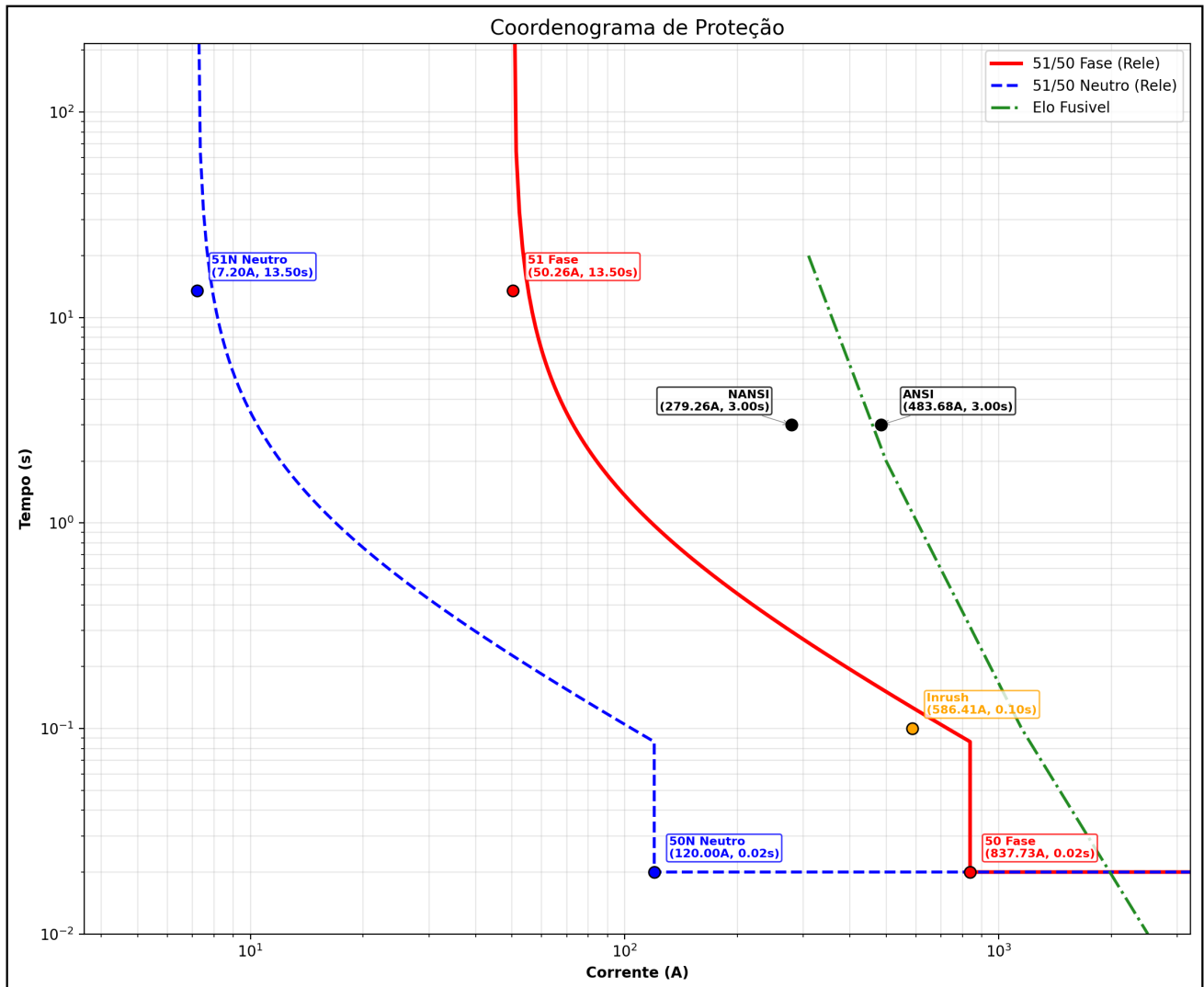
Fundamentals	Calculation Memory
Cable Impedance (4.0mm ² / 10m)	0.0555 Ohms
Relay Impedance (Nominal Burden)	(0.2 VA / 5 A) = 0.0080 Ohms
CT Secondary Resistance (25% of CT Burden)	(PF 0.9 / 5 VA) = 0.0500 Ohms
Z Burden (Cable + Relay + CT)	0.0555 + 0.0080 + 0.0500 = 0.1135 Ohms
Calculated Saturation Voltage [(Icc3 / CTR) x Z Burden]	69.81 x 0.1135 = 7.93 V
Nominal Saturation Voltage (CT)	12.00 V

4. Current Transformer Specification

Parameters	Nameplate Rating
Current Transformer Ratio / CTR	60:5 / 12
Accuracy Class / ALF	P / 15
Nameplate Marking (IEC Standard)	5VA 5P 15

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5. Protection Co-ordination Graph - ANSI 50/51



6. Relay Settings and Parameterization

Function	Phase Setting	Neutral Setting
IDMT (ANSI 51)	4.19A	0.60A
Instantaneous (ANSI 50)	69.81A	10.00A
Time Multiplier / IEC Curve	0.10 / VI - MUITO INVERSA	0.10 / VI - MUITO INVERSA

ADQUIRA AGORA - PAGAMENTO ÚNICO